
1. External APIs & Exchange Data
RBI Monetary Bulletins
NewsAPI & Financial RSS
BSE Directory & Corporate Actions
NSE Official EQUITY_L.csv
Yahoo Finance Live & Historical
Google Gemini API (gemini-3.8-flash)
2. Data Ingestion & Normalization
Winsorized Factor Z-Score Normalizer
Universe Sync & ISIN Resolver
Stale Price Filter (<=14 Days)
Liquidity Tiering (ADTV & Turnover)
3. PostgreSQL 16 Master (62 Tables)
economic_indicators & news
securities, listings, companies
market_prices & fundamentals
app_users & risk_assessments
4. Application Services & State Assembly
Demat Diagnosis Service
Candidate Universe Selector
PortfolioAgentState Assembly
5. LangGraph Agentic Intelligence
Suitability & Feasibility Node
Specialized Analysts (Macro, Mkt, Fund, News)
Lead Strategist Proposal
Independent Risk Critic (Mandate Audit)
Approved Candidates
6. Quantitative Optimization Engine
Deterministic SLSQP Optimizer
Whole-Share Integer Allocation
Residual Cash Conservation
Statutory Transaction Fees (STT, GST)
7. Fiduciary Validation & Audit
Critic Post-Optimizer Final Audit
Two-Level Institutional Explanation Engine
8. Client Delivery & Monitoring
ACID Portfolio Persistence
Market Hours Drift Monitor (09:15-15:30 IST)
Drift Alert
Next.js 16.4.2 Investor Dashboard